

Gemini Near Infra-Red Imager and Wavefront Sensor



OIWFS — Mechanism Testing

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niri_h ty_015 / 1.00 / 2001-07-02

This document describes the procedure for verifying that the OIWFS mechanical components are properly working.

1 Introduction

1.1 Purpose

This document describes the procedure for insuring that the OIWFS mechanical components are working properly. It assumes that all wiring has already been verified for correctness, and that the default channels are used for all input and output boards. It assumes that all of the boards have been correctly configured¹.

Note that although the Digital Input / Output board is intended to permit the OIWFS to interface to the interlock bus, there is no requirement that it actually must connect to the interlock bus. Support for the interlock bus is minimal, and is provided so that the user may define instrument-specific support, as required.

The current release of the OIWFS software requires a Bancomm board to interface to the A&G system. It can, however, be tested without the Bancomm board. The only operations that will fail are those that relate to the “follow” command.

Note that the OIWFS was initially designed to work with NIRI. Since delivery of NIRI, changes have been made to the NIRI software (including the OIWFS software). Later changes have not been propagated into the NIRI developers’ software, so some changes may need to be performed by the Gemini staff in order to integrate the OIWFS with the Gemini control system.

1. Note: For GNIRS, all boards except the Digital Input/Output board were correctly configured at the time that the system was shipped. The Digital Input/Output board should be configured according to Gemini standards, as described in [30].

1.2 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1.3 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

1.4 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

2 Precautions

2.1 Temperature

The mechanisms should never be operated while the instrument is being cooled or warmed. This is particularly important if the mechanism is being tested in a small liquid-nitrogen cooled dewar. Severe temperature gradients may occur, which can cause the mechanisms to bind. Operating under those circumstances can break the mechanism.

It is strongly recommended that a temperature sensor be placed on a part of the mechanism which is as far away from the source of heating or cooling, in order to verify that the entire mechanism has reached thermal equilibrium before operation. Note that for the focus stage, the moving part of the mechanism has very poor thermal contact with the rest of the mechanism. Also, the motor shaft has very poor thermal contact with the motor housing.

2.2 Motor Heating

The mechanisms are not intended to be operated continuously for extended periods of time (i.e., longer than five minutes.) During prolonged, continuous operation, the motors can become hot. This should be avoided. It is recommended that any tests which require continuous motion of the mechanisms should be performed intermittently, with adequate time to permit the motor to cool.

2.3 Wear

The mechanisms are not intended to be operated continuously. Avoid day-long tests, as this will reduce the lifetime of the mechanisms.

3 Procedure for Testing Mechanisms

The procedure described here assumes that the OIWFS control software (at least the Engineering subsystem) has already been configured and installed, as described in the release

notes. It assumes that the Engineering subsystem has been verified to work correctly in simulation mode.

3.1 Preparation

1. Source your local configuration file
2. Run niriEngStart. This will bring up the top level engineering screen.
3. Press the Interlocks button. This will bring up the interlocks screen.
4. Press the “WARM” button, to set the system temperature. (For cold tests, the “COLD” button should be used instead.) Wait for one minute. You must set the system temperature each time that the IOC is rebooted. You must set the temperature each time that the system completes warming or cooling. It is strongly recommended that the system temperature be set to “CHANGING” while the system is being warmed or cooled; this will set an interlock that will protect the mechanisms by keeping them from moving.
5. Close the Interlocks screen.
6. Return to the top level engineering screen.
7. Press the “Wavefront Sensor CC” button. This will bring up the top-level Wavefront Sensor Components screen.

4 Filter Wheel

1. From the top-level Wavefront Sensor Components screen, press the “Filter wheel” button. This will bring up the filter wheel engineering screen.
2. Identify the motor driver module which will be used to drive the filter wheel.
3. Enter a number in the new position (“New Pos”) field that is 40000 more than the current position (“Cur Pos”).
4. You should see all of the following things happen. If any of these does not, then there is a problem with the mechanism.
 5. The LED on the motor driver module should turn yellow.
 - If it is off, then the motor is not enabled
 - If it is green, then the motor is enabled, but not receiving pulses
 - If it is red, then the driver is in a fault state. Turn the motor driver box off for 20 seconds, then turn it back on again. If the LED again turns red, then there is a problem with the electronics, motor, or wiring.
6. The motor should immediately start moving.
7. The value of the sensors marked 1P and 1B should change slightly, but should be less than 0.100 V most of the time. Occasionally, the sensor values will rise up or drop down to a much larger value, and then immediately return back to a low value. It will take constant inspection of the meters to see this happen.
 - If all of these happen, then the motor is turning, the system is working at a minimal level, but is not necessarily performing to specification.
8. Press the Datum/Diag button. This will find the home position for the mechanism and immediately perform a quick mechanism check. Check the Status and Error/Info boxes. If all goes well, the system should return to the Idle state. There should be no messages in the Error/Info box at this time. If there are any messages, then the system has failed the test. See the calibration procedure.

9. If all goes well, press the “cycle” button. This will perform a continuous mechanism test. Twenty or thirty cycles should be sufficient to verify correct operation. End the cycle test by pressing the stop button.

4.1 Focus Stage

1. From the top-level Wavefront Sensor Components screen, press the “Focus stage” button. This will bring up the filter wheel engineering screen.
2. Locate the motor which is to drive the focus stage
3. By hand, with motor off, turn the main gear of the mechanism near the center of travel
4. Enter a number in the new position (“New Pos”) field that is 500 more than the current position (“Cur Pos”).
5. Verify that the motor turns and then stops.
6. Verify that the hall-effect sensor meters (labelled 2P and 2B) move.
7. Press the Datum/Diag button.
8. Watch and listen to the mechanism very carefully. If you can see or hear the mechanism hit a limit, stop it immediately. If this happens, there is a serious problem with the mechanism or its sensors. Low-level calibration may be required.
9. Press the Datum/Diag button. This will find the home position for the mechanism and immediately perform a quick mechanism check. Check the Status and Error/Info boxes. If all goes well, the system should return to the Idle state. There should be no messages in the Error/Info box at this time. If there are any messages, then the system has failed the test. See the calibration procedure.
10. If all goes well, press the “cycle” button. This will perform a continuous mechanism test. Twenty or thirty cycles should be sufficient to verify correct operation. End the cycle test by pressing the stop button.

4.2 Gimbal

Only the x-axis will be described here. The y-axis is tested identically.

1. From the top-level Wavefront Sensor Components screen, press the “Gimbal X” button. This will bring up the gimbal x-axis engineering screen.
2. Locate the motor which is to drive the filter wheel
3. Move the mechanism away from the limits of travel. (May require mechanical adjustment).
4. Enter a number in the new position (“New Pos”) field that is 70000 more than the current position (“Cur Pos”).
5. Verify that the motor turns and then stops.
6. Verify that the mirror support moves. Note that because of the high gear ratio, this motion is very difficult to see.
7. Verify that the two right hall-effect sensor meters (marked 2P and 2B) move.
8. Verify that the two left hall-effect sensor meters (marked 1P and 1B) move. Note that these meters will normally have a value that is close to 0. They will move off-zero very briefly. It takes careful observation to see this motion. If no motion is seen, repeat from step 4, with a value that is 70000 less than the starting position. If motion is still not seen, then the hall sensors are not being read properly or the entire gear train is not turning.

9. Press the Datum/Diag button. This will find the home position for the mechanism and immediately perform a quick mechanism check. Check the Status and Error/Info boxes. If all goes well, the system should return to the Idle state. There should be no messages in the Error/Info box at this time. If there are any messages, then the system has failed the test. See the calibration procedure.
10. If all goes well, press the “cycle” button. This will perform a continuous mechanism test. Because of the very large range of travel of this mechanism, two or three cycles should be sufficient to verify correct operation. End the cycle test by pressing the stop button.

